



Groundwater Quality Mapping in Parts of Kadapa and Anantapur Districts, Andhra Pradesh, India using Geographic Information System

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Abstract: Hydrogeochemistry of groundwater in parts of Kadapa and Anantapur districts, AP, India was used to assess the quality of groundwater for determining its suitability for drinking purpose. The area is located in the survey of India toposheet Number 57 J/3 lying between East 78°00'0" to 78°15'0" longitudes and 14°15'0" to 14°30'0" North latitudes covering an area of 720 sq. kms. Geologically, it is underlain mainly by Peninsular gneisses of Archean age followed by Gulcheru and Vemapalli formations comprising quartzites, conglomerates, dolomites and shales. Major geomorphic units are denudational hills, residual hills, pediments, pediplains, structural hills and valleys. The study area experiences a semiarid climate FOLLOWED BY water scarcity as well as water quality problems. Hence, a Geographic Information System (GIS) based assessment of spatiotemporal behaviour of groundwater quality has been carried out in the region. Twenty water samples were collected from twenty tube wells in the study area during pre and post monsoon seasons i.e April 2011 and September 2011. The parameters determined were: pH, total hardness (TH), chloride (Cl⁻), calcium (Ca⁺²) and fluoride (F⁻). Surface maps were prepared in the ARC- GIS 9.3 (spatial analyst module) to assess the quality in terms of spatial variation. The analysed parameters were in adherence to the desirable limits prescribed by WHO and Indian Standards for drinking water except fluoride (F⁻). High fluoride concentration was found in the villages covered with granitic gneissic complexes than the other rock formations. Weathering and leaching of fluorine-bearing minerals in rock formations under alkaline environment lead to the enrichment of fluoride in the groundwater. The spatial analysis of groundwater quality patterns of the study area shows seasonal fluctuations and these spatial patterns of physical and chemical constituents are useful in deciding water use strategies for various purposes.

Keywords: Hydrogeochemistry, Spatiotemporal, GIS, Southeastern part of Anantapur District, A.P

Introduction:

Water is the most valuable and vital resource for sustenance of life and also for any developmental activity. With the surface water sources dwindling to meet the various demands, groundwater has become the only reliable resource. The indiscriminate use of this vital natural resource is creating groundwater – mining problem in various parts of the world (Todd, 2007). Hence, the groundwater resource should be evaluated thoroughly, carefully and reliably on a real time basis to meet the ever growing needs. Groundwater is a natural resource for the economic and secure provision of drinking water without which human existence is unimaginable. However, the pressure on groundwater, in terms of both quantity and quality, has increased to an extent whereby not only drinking water sources but also sensitive ecosystems are threatened by contamination through human exploitation. It is a well-known fact that a polluted environment has a detrimental effect on health of people, animal life and

vegetation (Sujatha and Reddy 2003). Groundwater is extensively used for domestic, industrial and irrigation purposes. Water quality analysis is one of the most important issues in groundwater studies. Its monitoring and assessment is imperative for devising preventive measures against health hazards (Dhiman 2005). Quality of groundwater is equally important to its quantity owing to the suitability of water for various purposes. Variation of groundwater quality in an area is a function of physical and chemical parameters that are generally influenced by geological formations and anthropogenic activities (Subramani et al. 2005). The hydrogeochemical study reveals the zones and quality of water that are suitable for drinking, agricultural and industrial purposes (Anbazhagan and Archana 2004). The chemical quality of groundwater is related to the lithology of the area and the residence time of the water in contact with rock material. Weathered mantle, soils and atmosphere are the important factors responsible for contribution of dissolved solids to water.

It is impossible to control the dissolution of undesirable constituents in the waters after they enter the ground (Sreenivas Gowd 2005).

Geographical Information System (GIS) based groundwater studies were concentrated on the preparation of hydrogeomorphological maps, interpretation of lineaments and integrated terrain analysis (Kamaraju 1997; Srivastava and Bhattacharya 2000; Khan and Moharana 2002). Now GIS is increasingly being integrated with groundwater and surface water quality assessment models. GIS can be used as tool for spatial surveillance of epidemiological diseases and also as a spatial decision support system for public health management. The spatial patterns of chemical constituents are useful in deciding the water use strategies for various purposes (Collins et al 1998; Gogu et al, 2001; Dhiman 2005; Shankar et al., 2010). Assessing and monitoring the quality of groundwater is therefore, important to ensure sustainable safe use of these resources for the various purposes. In the study area, the data pertaining to the groundwater quality are not recorded till the time of this study. Hence the present study is an attempt to understand and identify the spatiotemporal behavior of groundwater quality of

southeastern part of Anantapur District, Andhra Pradesh.

Study Area:

The study area lies between Pulivendula and Sanivaripalli situated between parallels of $78^{\circ}00'0''$ to $78^{\circ}15'0''$ E longitude and $14^{\circ}15'0''$ to $14^{\circ}30'0''$ N latitude with intended boundary falling in Survey of India (SOI) topographic sheet # 57 J/03 covering an area of 720 sq. kms. The study area includes five revenue mandals namely, Pulivendula, Lingala, Udumakurti, Krishnamgaripalli, and Sanivaripalli and is shown in figure-1. The central and southern part of study area is occupied by high hills, ridges and valleys. The value of contours in the study area ranges from 200-600 m. The slope category is within 1-15 degrees. The Maddaleru is the only seasonal river fed by monsoon which is flowing in the NW direction in the southern part of the study area. The climate is dry with mean annual rainfall of 60 – 80 cm (Directorate of Economic and Statistics, Govt of AP) and mean annual temperature of 32°C . The month of May is considered to be the hottest (45°C) while December is considered to be the coldest (25°C).

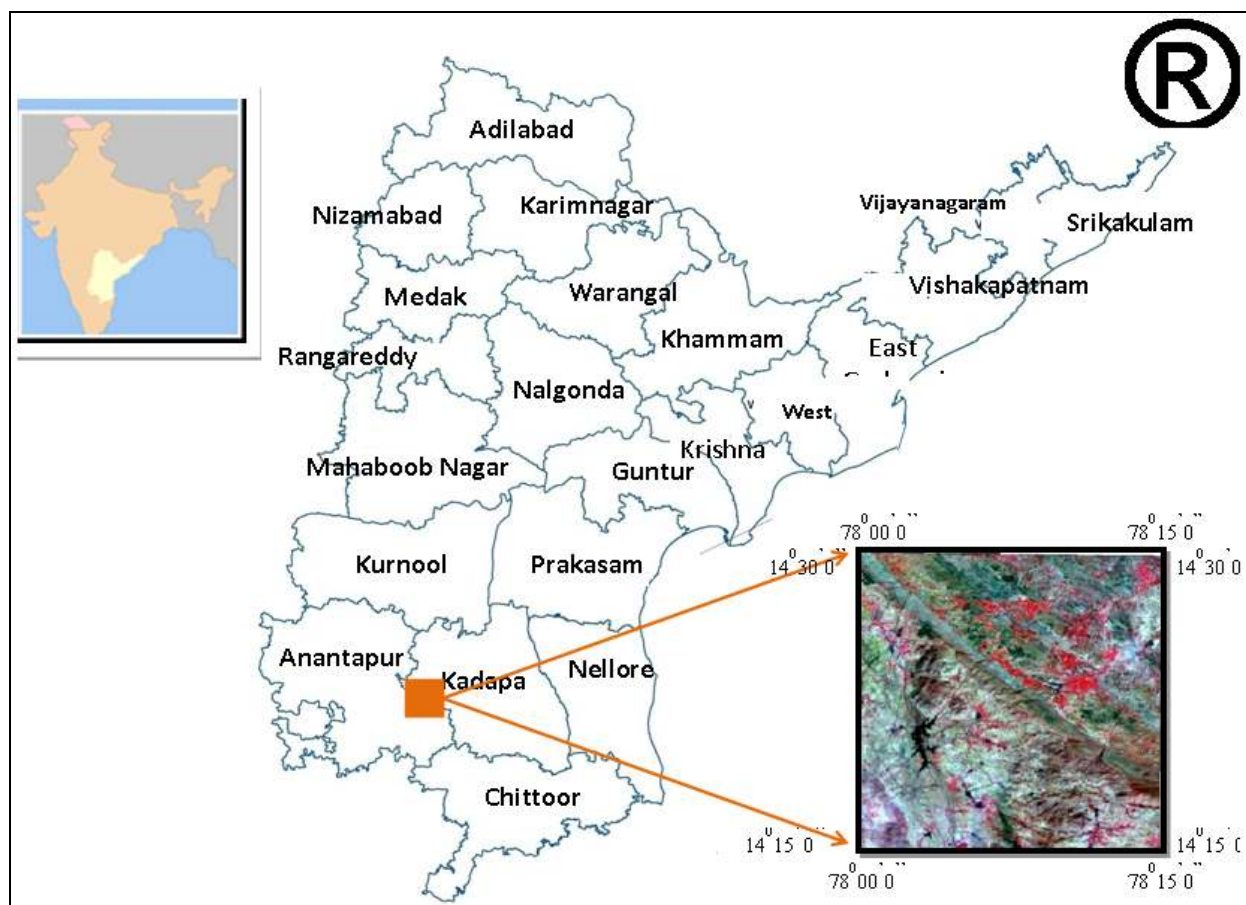


Figure 1: Study Area Location Map

Methodology:

Twenty water samples were collected from twenty tube wells in the study area during pre and post monsoon seasons i.e April 2011 and September 2011. The study is carried out with the help of topographic sheets, Garmin e Trex H (GPS) receiver, Arc GIS 9.3 and fieldwork. Toposheets were used to prepare the basemap, drainage map and to understand the general nature of the study area. GPS is used to map the location of each sampling well and finally the results were taken to the GIS for further analysis. The field work included water level measurements, well inventory and collection of water samples from tube wells and the study of geological and geomorphological features of the area in general. The groundwater samples collected during June 2011 were analyzed, as per the procedure of APHA (1995), and suggested precautions were taken to avoid contamination. The various parameters determined were: pH, total hardness (TH), chloride (Cl^-), calcium (Ca^{2+}) and fluoride (F^-). pH was determined by pH meter; TH, Cl^- , Ca^{2+} by titrimetry; F^- was determined by using ion selective electrode (Orion 4 star ion meter, Model: pH/ISE). The location of each well was taken into the GIS environment and the results of each parameters analysed were added to the concerned wells. Spatial analyst, an extended module of ARC-GIS 9.3 was used to find out the spatiotemporal behavior of the groundwater quality parameters.

Results and Discussion:

Chemical Analysis:

The results of chemical and statistical parameters analysed are given in Table 1. Understanding the quality of groundwater is as important because it is the main factor determining its suitability for drinking, domestic, agricultural and industrial purposes. The pH values of groundwater range from 7.1 to 8.2 with mean of 7.63 during April 2011. However during September 2011, it ranges from 6.4 to 8 with a mean of 7.61. This shows that the groundwater of the study area is alkaline. The total hardness values range from 60 to 180 mg/l with a mean of 124 mg/l during pre monsoon season and from 50 to 170 mg/l with a mean of 114 mg/l during post monsoon season September 2011. Calcium (Ca^{2+}) in the groundwater varies from 8 to 992 mg/l with a mean of 119.6 mg/l during pre monsoon season and from 16 to 900 mg/l with a mean of 110.4 mg/l during post monsoon season. Chlorides ranged between 17.7 mg/l to 166.8 mg/l with a mean of 82.69 mg/l during pre monsoon and from 17.7 to 166 mg/l with a mean of 80.99 mg/l during post monsoon season. The fluoride values ranged from 0.1 to 6.7 mg/l with a mean of 1.695 mg/l during pre monsoon season and from 0.1 to 6.0 with a mean of 1.575 mg/l during post monsoon season. The analytical results of physical and chemical

parameters of groundwater were compared with the standard guideline values as prescribed by the World Health Organization (WHO 1983) and Indian Standards (ISI 1983) for drinking and public health purposes (Table 2). The parameters analyzed were above the desirable limits of both standards.

Spatial Analysis of Groundwater Quality:

The GIS based analysis of spatiotemporal behavior of the groundwater quality in the study area was done using the spatial analyst module of Arc GIS 9.3. Powerful spatial analysis is feasible once the database is established. The interpolation technique used in the analysis is inverse distance weighted (IDW) method. IDW is an algorithm for spatially interpolating or estimating values between measurements. Each value estimated in an IDW interpolation is a weighted average of the surrounding sample points. Weights are computed by taking the inverse of the distance from an observations location to the location of the point being estimated (Burrough and MC Donnell 1998). The inverse distance can be raised to a power (e.g. linear, squared, cubed) to model different geometries (e.g. line, area, volume) (Guan et al., 1999). In a comparison of several different deterministic interpolation procedures, Burrough and Mc Donnell (1998) and Mathes et al. (2006) found that using IDW with a squared distance term yielded results most consistent with original input data. This method is suitable for datasets where the maximum and minimum values in the interpolated surface commonly occur at sample points (ESRI 2002).

The spatial pattern of parameters was analysed in two seasons, i.e., monsoon and post monsoon seasons. The spatial analysis of pH during the two seasons shows the change in the areas of spatial highs and lows (fig: 2). During April 2011 the spatial highs are showing a clear NW-SE alignment but in September 2011 it shows an eastward shift with a spatial maximum in the northeastern side. The spatial distribution of total hardness (TH) highs are very prominent in post monsoon when compared to pre monsoon season (Fig: 3). This hardness may be attributed to the presence of carbonates in the form of dolomites in the study area. Most of the study area has moderate distribution of calcium (Ca^{2+}) except at Yogi Vemana Reservoir as shown as Isolated highs (fig: 4). The Isolated patches of spatial highs of chlorides (Cl^-) are more prominent at Chinakuddala, Ippatla and Mallepalli and Sanivaripalli villages in both pre and post monsoon seasons (Fig: 5). Southwestern part of the study area is showing higher concentration of fluorides (fig: 6). High fluoride concentration was found in the villages covered with granitic gneissic complexes than the other rock formations. Weathering and leaching of fluorine-bearing minerals in rock formations under alkaline environment

lead to the enrichment of fluoride in the groundwaters. Among the twenty groundwater samples analysed in the study area, 50% of samples had high fluoride content than the maximum permissible limit (1.5 mg/l). A high rate of evapotranspiration, comparatively low rainfall,

intensive irrigation, alkaline environment, longer residence time of water in the weathered aquifer zone and low rate of dilution are favourable factors for the dissolution of fluorine- bearing minerals and thereby increase of fluoride concentration in groundwater.

Table 1: Result of Chemical Analysis of Groundwater Samples Collected from the Study Area during Pre and Post Monsoon Seasons.

S.No	Village name	p ^H		TH		Ca ⁺²		Cl ⁻		F ⁻	
		Pre	Post	mg/l		mg/l		mg/l		mg/l	
				Pre	Post	Pre	Post	Pre	Post	Pre	Post
1	Kottala	7.9	7.9	100	100	104	100	31.9	30	0.2	0.2
2	Gajjalapalli - 1	7.8	7.4	140	120	88	80	128	120	1.5	1.2
3	Gajjalapalli - 2	7.4	7.4	160	150	16	16	110	110	6.7	6
4	Gajjalapalli - 3	7.6	7.6	100	100	8	10	92.3	90	4	4
5	malakavemula	8.2	8	80	70	32	22	56.8	56	1.9	1.7
6	sanevaripalli	8	8	140	120	48	40	131	131	1.7	1.7
7	kotareddypalli	7.9	7.9	100	100	16	16	67.4	67.4	3.8	3.5
8	Vaddekindi tanda	8.2	8	100	90	240	220	63.9	64	2.7	2.3
9	mallepalli	7.1	7.4	180	160	64	64	121	120	2.9	2.6
10	Yogi Vemana reservoir	8	8	120	100	992	900	74.5	74.5	2.4	2.2
11	dorigallu	7.2	7.2	100	100	32	30	88.7	90	2.1	2.1
12	eguvapalli	6.9	7.2	140	120	256	250	42.6	42.6	0.1	0.1
13	diguwapalli	7.2	7.2	60	50	32	30	17.7	17.7	0.4	0.4
14	ambakapalli	6.1	6.4	120	110	64	60	28.4	28.4	0.6	0.6
15	Nalagondavaripalli- 1	7.9	7.9	80	80	104	90	63.9	63.9	0.7	0.7
16	Nalagondavaripalli- 2	7.8	7.8	120	120	88	80	78.1	78	0.2	0.2
17	mallikarjunapuram	8	8	100	90	80	80	28.4	28.4	0.7	0.7
18	ippatla	7.8	7.8	180	160	32	30	160	140	0.5	0.5
19	chinnakuddala	7.6	7.2	180	170	40	40	167	166	0.5	0.5
20	lingala	8	8	180	170	56	50	103	102	0.3	0.3
Maximum		8.2	8	180	170	992	900	167	166	6.7	6
Minimum		7.1	6.4	60	50	8	16	17.7	17.7	0.1	0.1
MEAN VALUE OF PARAMETERS		7.6	7.61	124	114	120	110	82.7	81	1.7	1.58
STANDARD DEVIATION		0.5	0.42	37	33.4	216	196	43.6	41.5	1.7	1.54

Table 2: A Comparison of Water Quality Parameters (Drinking) of the Study Area with Whom and Indian Standards

Parameters	WHO International Standard (1983)	Indian Standards for drinking water specification	mean value of parameters under analysis	
			Pre	Post
pH	7-8.5	6.5-8.5	7.63	7.61
TH (mg/l)	100	300	124	114
Ca ⁺² (mg/l)	200	75	119.6	110.4
Cl ⁻ (mg/l)	200	250	82.69	80.99
F ⁻ (mg/l)	1.5	1.5	1.695	1.575

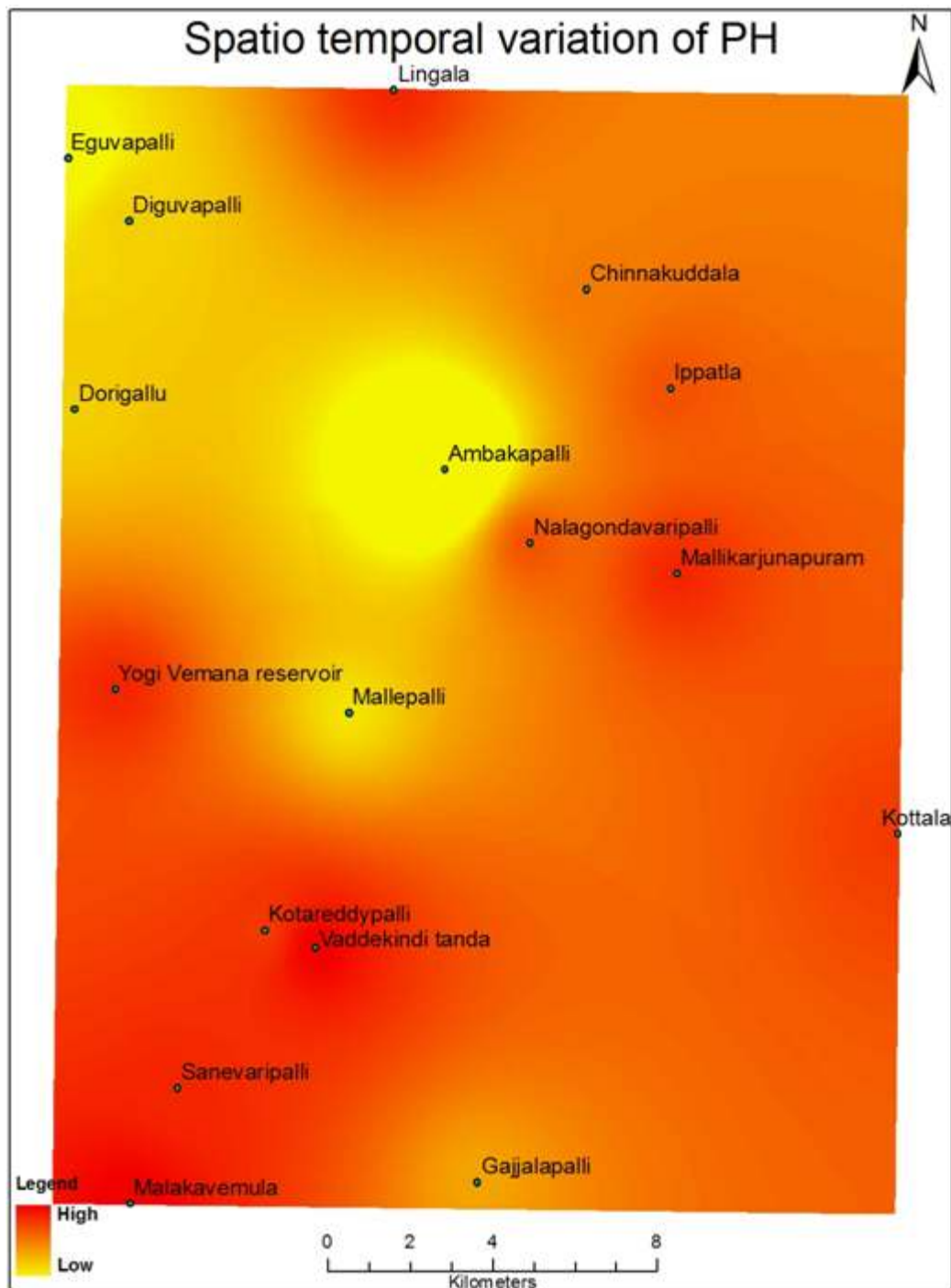


Figure 2: Map Showing Spatiotemporal Variation of pH in the Study Area

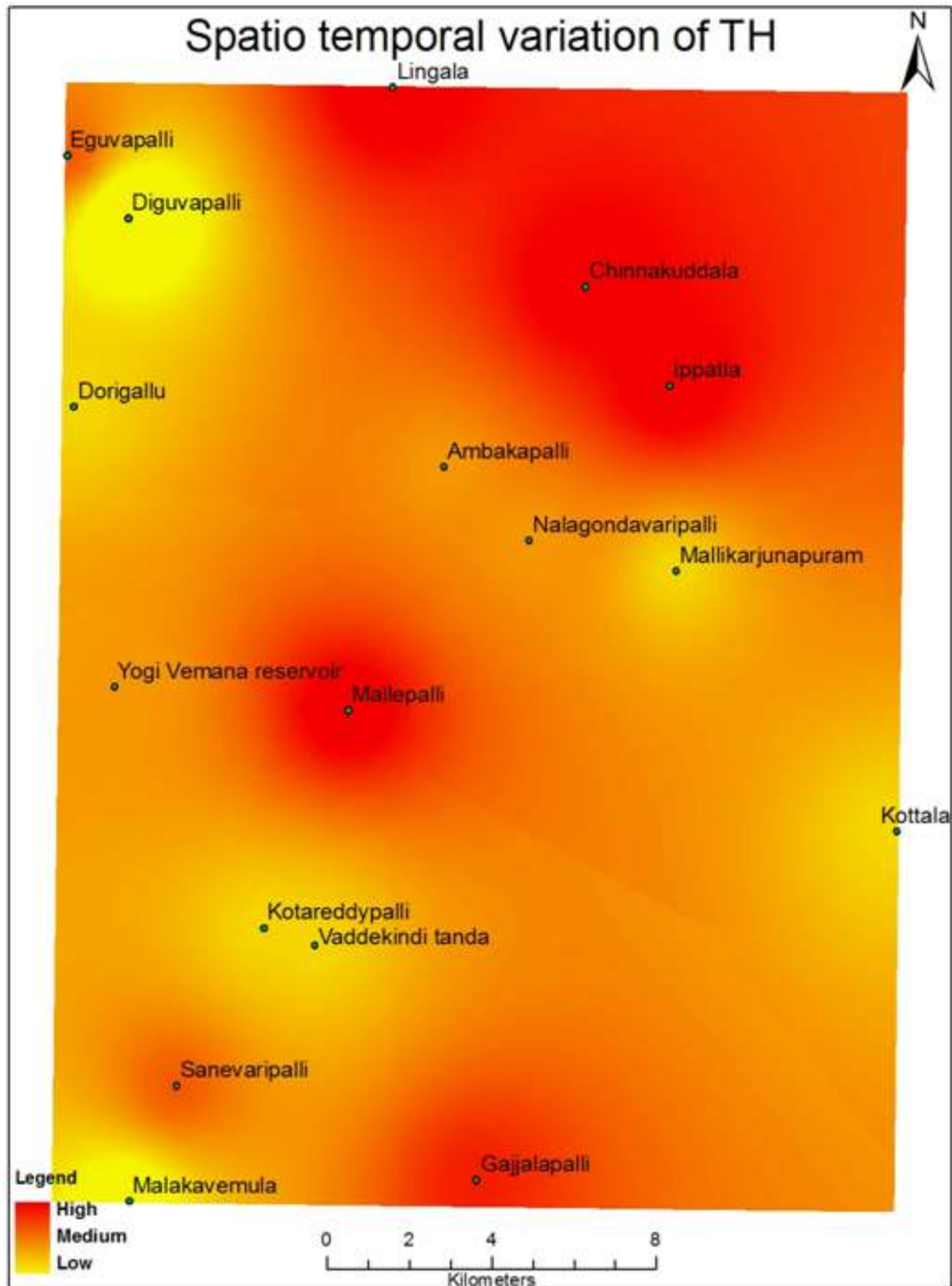


Figure 3: Map Showing Spatiotemporal Variation of TH in the Study Area

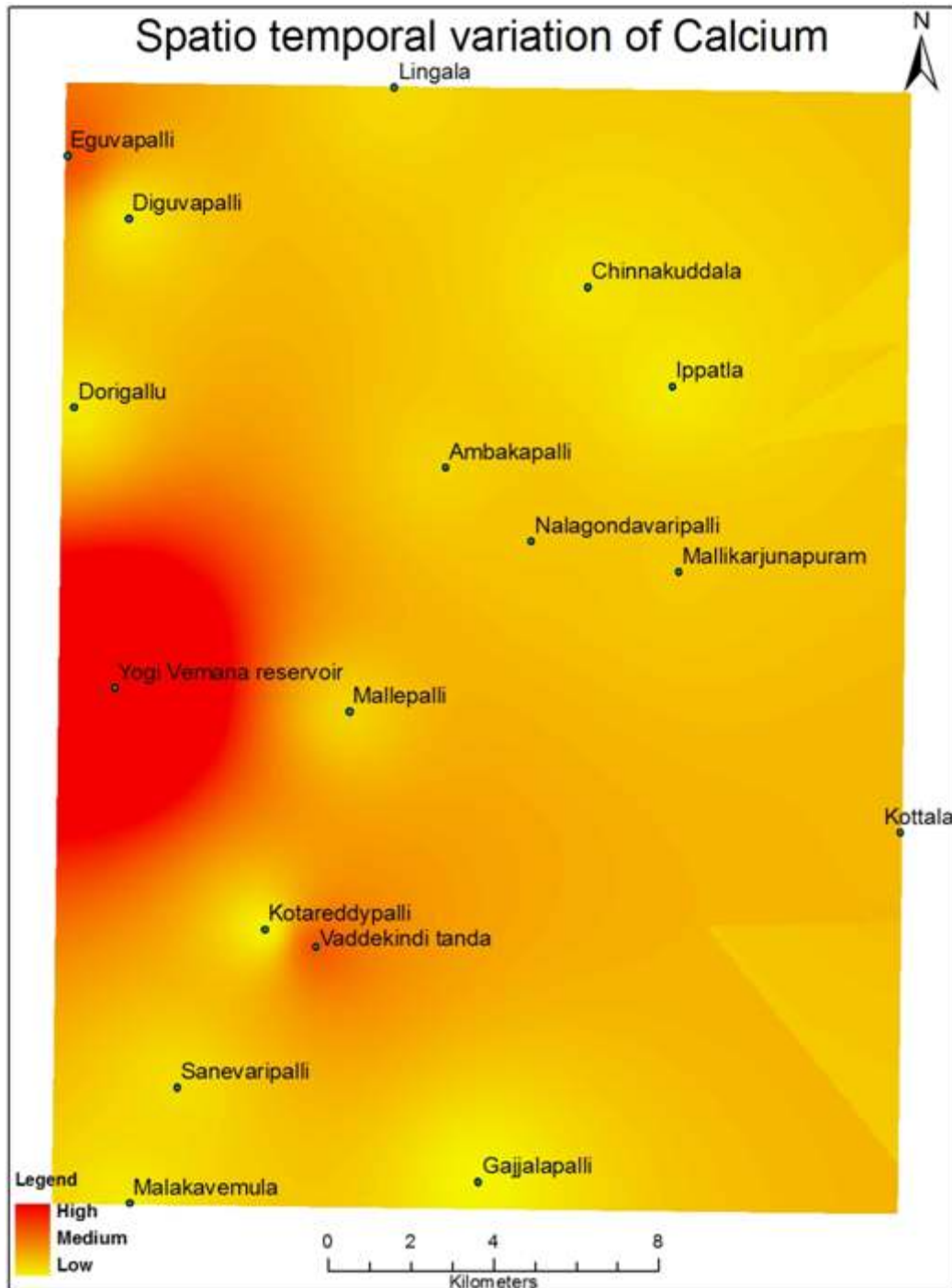


Figure 4: Map Showing Spatiotemporal Variation of Ca^+ in the Study Area

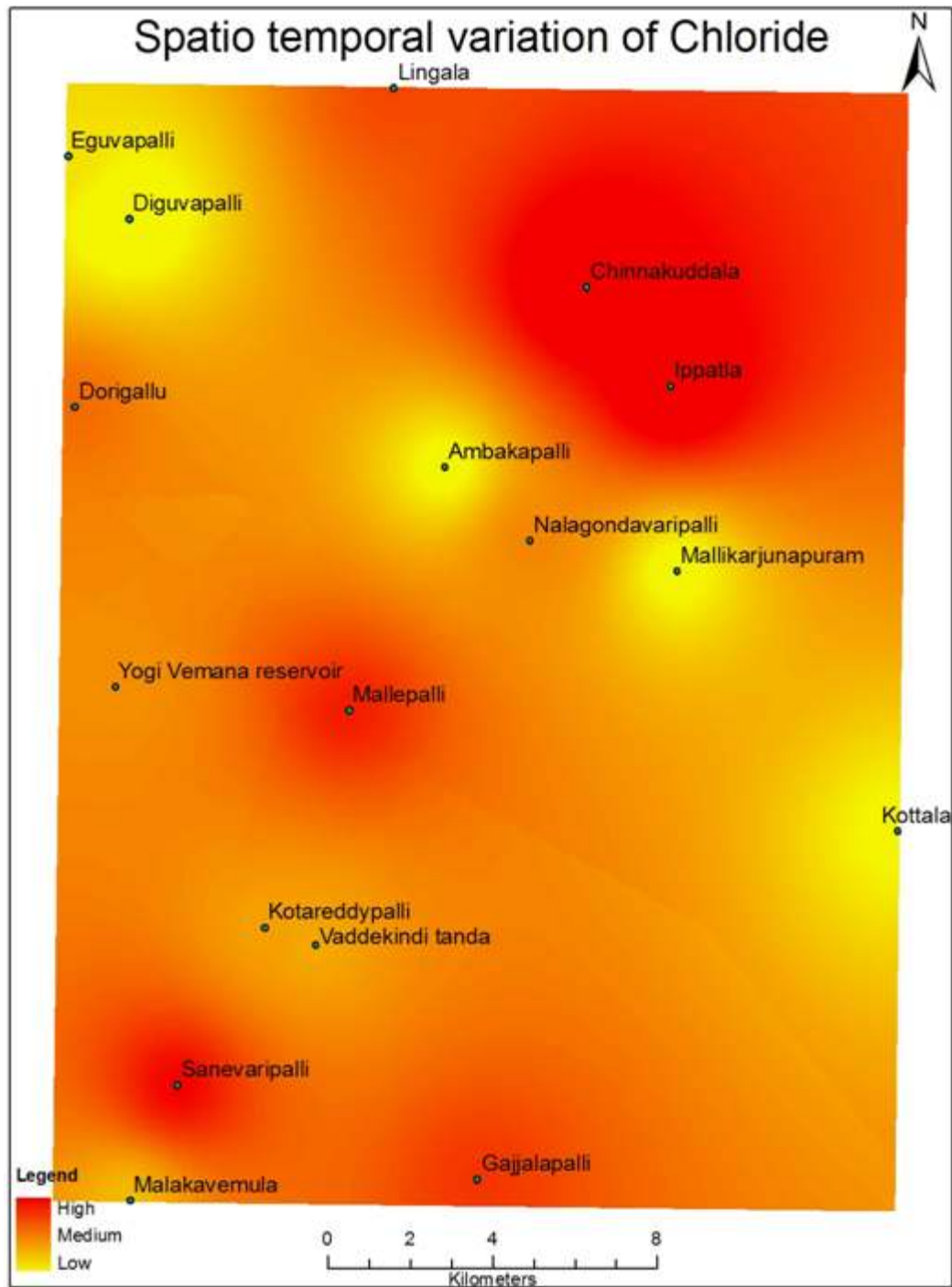


Figure 5: Map Showing Spatiotemporal Variation of Cl^- in the Study Area

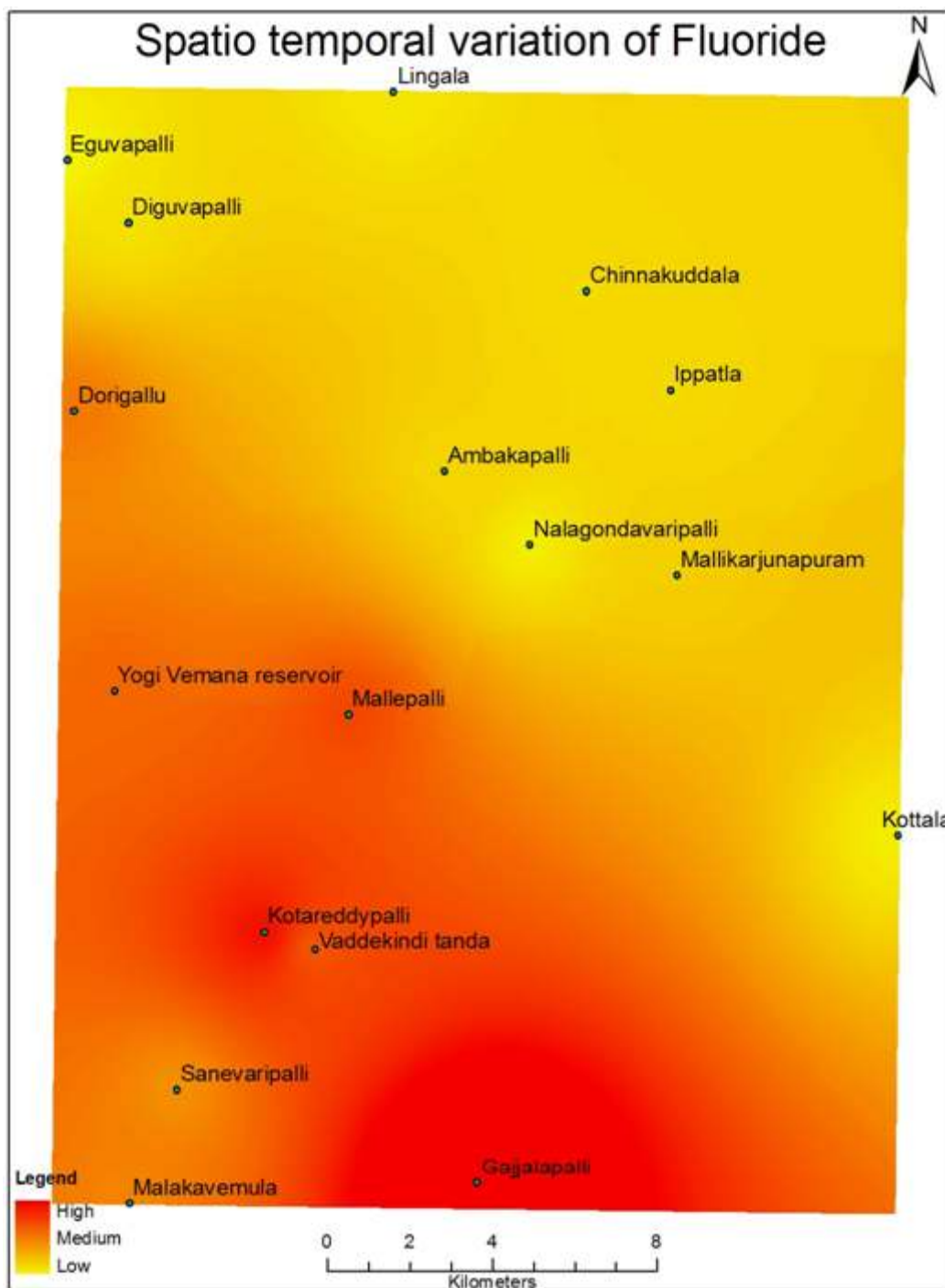


Figure 6: Map Showing Spatiotemporal Variation of F^- in the Study Area

Conclusion:

The purpose of the present study is to understand the groundwater quality in the study area and pictorially represent it using GIS. Most of the chemical constituents exclusive of fluorides in the water samples collected from the study area comply with water quality standards. The chemical characteristics of the water samples reveal slight seasonal variations. The waters of

the region are alkaline. The spatial distribution of total hardness (TH) highs are very prominent in post monsoon when compared to pre monsoon season (Fig: 3). This hardness may be attributed to the presence of carbonates in the form of dolomites in the study area. The Isolated patches of spatial highs of chlorides (Cl^-) are more prominent at Chinakuddala, Ippatla and Mallepalli and Sanivaripalli villages in both pre and

post monsoon seasons. High fluoride concentration is found in the villages covered with granitic gneissic complexes than the other rock formations. Weathering and leaching of fluorine-bearing minerals in rock formations under alkaline environment lead to the enrichment of fluoride in the groundwater.

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